AHBA–GTEx Imputation by Deterministic and Probabilistic Latent Alignment

1 Introduction

The Allen Human Brain Atlas (AHBA) and GTEx provide complementary but structurally mismatched views of human brain transcriptomics. AHBA supplies dense atlas-level measurements over a canonical parcellation, whereas GTEx provides subject-specific observations over only a sparse subset of regions [Hawrylycz et al.(2012)]. GTEx broadens population coverage and subject diversity, but only over a limited set of brain regions [GTEx Consortium(2013), GTEx Consortium(2017), GTEx Consortium(2020)]. Our aim is to impute, for each GTEx subject, a complete parcel-by-gene expression matrix over all canonical parcels by leveraging the shared spatial structure encoded in the AHBA atlas. This is not a standard missing-data problem: it is cross-platform missing-region inference under domain shift, because the atlas reference is derived from microarray measurements (AHBA) while subject-level observations in GTEx arise from bulk RNA sequencing.

A common assumption in this setting is that inter-regional transcriptomic variation is dominated by a low-dimensional spatial structure that is more completely visible in the atlas than in any individual GTEx subject. The central modeling question is therefore how to infer that structure, align sparse subject measurements to the atlas frame, and extend subject-specific expression over unobserved parcels. Two complementary approaches are developed here. The first is the deterministic latent alignment model (DLAM), based on subject-specific partial least squares (PLS), affine latent alignment, and spatial interpolation. The second is the probabilistic latent alignment model (PLAM), in which the atlas defines a global factor structure, each subject is represented by an orthogonally aligned latent deviation field, and unobserved parcels are inferred by Gaussian-process conditioning.

2 Data Setting and Notation

Let $R = 150$ denote the canonical parcel set and G the number of genes under analysis. AHBA defines a dense parcel-by-gene atlas

$$X_A \in \mathbb{R}^{R\times G},$$

while subject s contributes GTEx observations on a sparse subset of parcels

$$\Omega_s \subset \{1,\dots,R\}, \qquad X_s^{(G)} \in \mathbb{R}^{|\Omega_s|\times G}.$$

All spatial quantities are defined with respect to canonical parcel centroids

$$Y \in \mathbb{R}^{R\times 3}.$$

The latent dimension is denoted by K and fixed to $K = 3$ throughout. Atlas-scale heatmaps and scatter diagnostics are shown on the matched high-variance subset with $G = 88$ for readability, whereas leave-one-region-out diagnostics are computed over the full gene set so that the cross-validated evaluation remains aligned with transcriptome-wide completion.

3 Cross-Dataset Harmonization

Both alignment methods begin with the same preprocessing step: AHBA and GTEx are mapped into a common harmonized expression domain before any latent alignment is attempted. The purpose of harmonization is to remove systematic, platform-induced differences in marginal gene-expression distributions while preserving the relative biological variation across parcels and subjects.

3.1 General harmonization model

Let $\mathcal{H}_A$ and $\mathcal{H}_G$ be platform-specific transformation functions parameterized by shared parameters θ_H, satisfying the following two properties: (i) the mapped expressions are on a common scale, in the sense that $\mathcal{H}_A(X_A;\theta_H)$ and $\mathcal{H}_G(X_s^{(G)};\theta_H)$ have compatible marginal distributions across genes; and (ii) biologically informative between-parcel and between-subject contrasts are preserved under the transformations. The shared preprocessing transformation is written as

$$X_A^{(h)} = \mathcal{H}_A(X_A;\theta_H), \qquad X_{s,h}^{obs} = \mathcal{H}_G(X_s^{(G)};\theta_H). \tag{3.1}$$

The parameters θ_H are estimated using the AHBA atlas together with the available GTEx rows, treating the two platforms as two batches. Both DLAM and PLAM receive the harmonized atlas $X_A^{(h)}$ and harmonized subject observations $X_{s,h}^{obs}$ as input, and both ultimately produce a completed harmonized subject atlas

$$\hat{X}_s^{(h)} \in \mathbb{R}^{R\times G}.$$

3.2 Implementation via ComBat-style Harmonization

In the default pipeline, Eq. (3.1) is implemented by a ComBat-style batch harmonizer fit jointly to AHBA and eligible GTEx rows [Johnson et al.(2007)]. We follow the notation of Johnson et al. and write y_{ijg} for the observed expression of gene g in sample j from batch i. In our setting, the two batches are $i = A$ for AHBA and $i = G$ for GTEx. ComBat models each gene as

$$y_{ijg} = \alpha_g + \mathbf{x}_{ij}^\top \boldsymbol{\beta}_g + \gamma_{ig} + \delta_{ig}\varepsilon_{ijg}, \tag{3.2}$$

where α_g is the gene-specific intercept, $\mathbf{x}_{ij}$ is a vector of row-level covariates, $\boldsymbol{\beta}_g$ gives the gene-specific covariate effects, γ_{ig} is the additive batch effect, and δ_{ig} is the multiplicative batch scale. In the default analysis, $\mathbf{x}_{ij}$ contains normalized age, a sex indicator, and broad anatomical macro-system indicators (visual-somatomotor cortex, association cortex, subcortical, cerebellar). With only two batches and AHBA contributing a small post-mortem cohort, a single pooled OLS estimate of $\boldsymbol{\beta}_g$ across both batches is unreliable for any covariate whose distribution correlates with batch identity: at this sample size, systematic demographic differences between AHBA and GTEx can dominate the pooled fit and bias $\boldsymbol{\beta}_g$ away from either batch's true effect. Our default pipeline therefore estimates $\boldsymbol{\beta}_g^{(i)}$ per batch for the subject-level demographic covariates (age, sex), and retains a single pooled coefficient for the parcel-level macro-system indicators, which has documented shared variance structure across batches. With a larger number of batches and more diverse per-batch sampling, a fully pooled covariate model would be appropriate; in the present two-batch, small-AHBA setting, the mixed per-batch / pooled formulation is the principled default.

Operationally, the pooled AHBA–GTEx expression table is first adjusted for the fitted covariate contribution $\mathbf{x}_{ij}^{\top}\hat{\boldsymbol{\beta}}_g^{(i)}$. For each gene, the covariate-adjusted values are centered and scaled using pooled statistics across both platforms:

$$s_{ijg} = \frac{y_{ijg} - \mathbf{x}_{ij}^{\top}\hat{\boldsymbol{\beta}}_g^{(i)} - \hat{\alpha}_g}{\hat{\sigma}_g}.$$

Within this standardized space, platform effects are estimated separately for AHBA and GTEx. The raw ComBat estimates are the batch-specific mean and variance of s_{ijg},

$$\hat{\gamma}_{ig} = \frac{1}{n_i}\sum_{j=1}^{n_i} s_{ijg}, \qquad \hat{\delta}_{ig} = \mathrm{Var}_{j=1,\dots,n_i}(s_{ijg}),$$

where n_i is the number of rows in batch i. Following the empirical-Bayes motivation of ComBat, these gene-wise batch estimates are stabilized by shrinkage across genes, yielding posterior batch adjustments $\hat{\gamma}_{ig}^*$ and $\hat{\delta}_{ig}^*$. In our implementation, this shrinkage is a sample-size weighted average between the gene-specific batch estimate and the corresponding batch-wide average across genes, with variance shrinkage performed on the log-variance scale. The standardized expression is then corrected by removing the estimated additive shift and multiplicative scale,

$$s_{ijg}^{adj} = \frac{s_{ijg} - \hat{\gamma}_{ig}^*}{\sqrt{\hat{\delta}_{ig}^*}}.$$

The corrected expression is reconstructed on the original gene scale while restoring the fitted covariate contribution:

$$y_{ijg}^{corr} = s_{ijg}^{adj}\hat{\sigma}_g + \hat{\alpha}_g + \mathbf{x}_{ij}^{\top}\hat{\boldsymbol{\beta}}_g^{(i)}.$$

Because AHBA and GTEx do not sample the same biological object, the implementation adds a final overlap-based calibration after ComBat correction. For each gene, corrected AHBA and GTEx values are aggregated to parcels observed in both datasets, and a gene-wise affine map is fit from corrected GTEx parcel means to corrected AHBA parcel means:

$$\bar{y}_{Arg}^{corr} \approx a_g \bar{y}_{Grg}^{corr} + d_g,$$

where r indexes shared parcels. The final harmonized values are therefore

$$y_{ijg}^{(h)} = \begin{cases} y_{ijg}^{corr}, & i = A, \\ a_g y_{ijg}^{corr} + d_g, & i = G. \end{cases}$$

Thus AHBA defines the reference harmonized atlas scale, while GTEx observations are first ComBat-corrected and then calibrated onto that reference scale. The resulting $X_A^{(h)}$ and $X_{s,h}^{obs}$ are the shared inputs used by the naive atlas-fill baseline, DLAM, and PLAM.

3.3 Naive Atlas-Fill Baseline

The same harmonized domain also supports a simple atlas-prior baseline. For a GTEx subject s, the naive fill predictor preserves all observed harmonized parcels and fills every unobserved parcel directly from the harmonized AHBA atlas:

$$\hat{x}_{srg}^{\text{naive}} = x_{A,rg}^{(h)}, \qquad r \notin \Omega_s, \tag{3.3}$$

$$\hat{x}_{srg}^{\text{naive}} = x_{srg}^{(G,h)}, \qquad r \in \Omega_s. \tag{3.4}$$

This baseline is not an alignment model: it contains no subject-specific latent structure beyond the exact preservation of observed parcels. It serves as a harmonized atlas-prior reference for both quantitative benchmarking and residual visualization.

4 Deterministic Latent Alignment Model (DLAM)

4.1 Intuition

DLAM treats the observed parcels of a subject as sufficient to estimate a subject-specific latent expression basis that is spatially structured. That basis is aligned to an atlas reference basis by an affine change of coordinates. After the subject-specific latent spatial trajectory has been transported into the atlas frame, it is interpolated over unobserved parcels and decoded back into gene space. The procedure is modular: harmonization, latent decomposition, affine alignment, interpolation, ridge regression, and decoding are estimated sequentially as point estimates.

4.2 Joint PLS decomposition

The key idea of DLAM is to factorize the subject's harmonized expression matrix $X_{s,h}^{obs}$ into two components: one that is specifically aligned with the spatial distribution of the observed parcels, and one that captures the residual gene-expression variation. This is achieved by fitting a joint partial least squares (PLS) decomposition that simultaneously decomposes expression and spatial covariates, finding latent directions of maximum covariance between the two.

Formally, let $\tilde{X} = X_{s,h}^{obs}$ and $\tilde{Y} = Y_{\Omega_s}$ denote the centered expression and spatial matrices at the observed parcels. PLS finds successive pairs of weight vectors $(p_k,\, c_k)$ with $p_k \in \mathbb{R}^G$ and $c_k \in \mathbb{R}^3$ solving

$$(p_k, c_k) = \arg \max_{\|p\|=\|c\|=1} \operatorname{Cov}\Big(\tilde{X}\, p,\ \tilde{Y}\, c\Big), \tag{4.1}$$

deflating the matrices between components. The resulting K component pairs yield latent score matrices

$$X_{s,h}^{obs} \approx Z_s W_s^\top, \qquad Z_s \in \mathbb{R}^{|\Omega_s| \times K},\ W_s \in \mathbb{R}^{G \times K}, \tag{4.2}$$

$$Y_{\Omega_s} \approx S_s C_s^\top, \qquad S_s \in \mathbb{R}^{|\Omega_s| \times K},\ C_s \in \mathbb{R}^{3 \times K}, \tag{4.3}$$

where Z_s are latent expression scores and S_s are latent spatial scores. By construction of the PLS criterion, each column $z_{s,\cdot,k}$ of Z_s is the linear combination of genes that maximally co-varies with the corresponding spatial score $s_{s,\cdot,k}$. This means Z_s captures precisely the spatially-structured component of expression: it is the part of $X_{s,h}^{obs}$ that can be predicted from spatial location.

The atlas reference decomposition is obtained analogously on $X_A^{(h)}$:

$$X_A^{(h)} \approx Z_A^{det} \Big(W_A^{det}\Big)^\top, \qquad Z_A^{det} \in \mathbb{R}^{R \times K},\ W_A^{det} \in \mathbb{R}^{G \times K}. \tag{4.4}$$

4.3 Affine alignment, interpolation, and decoding

Because the subject and atlas PLS decompositions are fit independently, the resulting latent spaces are in different coordinate systems. The subject latent expression scores Z_s are aligned to the atlas reference scores Z_A^{det} on the observed parcels by solving a regularized affine regression:

$$(B_s, b_s) = \arg \min_{B,b} \Big\| Z_s B + \mathbf{1} b^\top - Z_{A,\Omega_s}^{det} \Big\|_F^2 + \lambda_B \|B\|_F^2. \tag{4.5}$$

The fitted affine map is then applied to both the latent expression and spatial scores:

$$Z'_s = Z_s B_s + \mathbf{1} b_s^\top, \tag{4.6}$$

$$S'_s = S_s B_s. \tag{4.7}$$

Why the same B_s applies to both Z_s and S_s. The PLS decomposition in Eqs. (4.2)–(4.3) operates in a *shared* latent space: both Z_s and S_s are projections of their respective data matrices onto the same set of K latent directions, chosen to maximize cross-covariance. Concretely, one can write $z_{s,r} = X_{s,r}^{(h)} W_s$ and $s_{s,r} = y_{\Omega_s,r} C_s$, where the loading directions W_s and C_s are jointly determined by the PLS criterion (4.1). The alignment matrix B_s represents a linear change of basis in this shared PLS latent space. Applying it consistently to Z_s and S_s is therefore necessary to maintain the correspondence between expression modes and spatial modes that was established by PLS: if the expression basis is rotated without rotating the spatial basis, the relationship between spatial position and predicted expression is lost.

The aligned latent spatial coordinates are extended to all parcels by an RBF interpolator

$$\hat{S}'_s(r) = f_s(y_r), \qquad r = 1, \ldots, R, \tag{4.8}$$

after which a ridge regression maps interpolated latent spatial coordinates back to aligned latent expression coordinates,

$$\hat{Z}'_s = \hat{S}'_s M_s + \mathbf{1} \beta_s^\top. \tag{4.9}$$

The aligned coordinates are then returned to the subject-specific score system,

$$\hat{Z}_s = \left(\hat{Z}'_s - \mathbf{1} b_s^\top \right) B_s^{-1}, \tag{4.10}$$

and decoded into harmonized expression via the subject-specific gene loadings,

$$\hat{X}_{s,h} = \hat{Z}_s W_s^\top. \tag{4.11}$$

Observed parcels are preserved exactly by overwrite,

$$\hat{x}_{srg}^{(h)} = x_{srg}^{(G,h)}, \qquad r \in \Omega_s. \tag{4.12}$$

The deterministic implementation additionally restricts anatomically implausible borrowing during subject-to-parcel assignment by preferring anchors in the same hemisphere and brain area before falling back to purely distance-based matches.

4.4 Initialization

Initialization is direct because each stage is either closed-form or a standard regression problem. The shared harmonizer in Eq. (3.1) is fit first. The atlas reference decomposition in Eq. (4.4) and the subject-specific PLS decomposition in Eqs. (4.2)–(4.3) are then fit from the harmonized matrices. The affine parameters (B_s, b_s) are initialized by least squares on the overlap objective in Eq. (4.5). The RBF interpolator in Eq. (4.8) and the ridge regression in Eq. (4.9) are fit on the observed parcels only. No latent state is propagated across subjects.

Algorithm 1 Deterministic alignment for subject s

Require: Atlas expression X_A, subject observations $X_s^{(G)}$ on Ω_s, parcel centroids Y, latent dimension K

Ensure: Completed harmonized atlas $\hat{X}_{s,h}$

1: Fit the shared harmonizer and compute $X_A^{(h)}$ and $X_{s,h}^{obs}$ using Eq. (3.1)
2: Fit the atlas reference latent decomposition in Eq. (4.4)
3: Fit the subject-specific joint PLS decompositions in Eqs. (4.2) and (4.3)
4: Estimate (B_s, b_s) by the affine alignment objective in Eq. (4.5)
5: Compute aligned subject coordinates by Eqs. (4.6) and (4.7)
6: Fit the RBF interpolator and evaluate $\hat{S}'_s(r)$ for all parcels using Eq. (4.8)
7: Fit ridge regression from observed S'_s to observed Z'_s and predict $\hat{Z}'_s$ using Eq. (4.9)
8: Map predicted aligned coordinates back to the subject-specific system with Eq. (4.10)
9: Decode to harmonized expression using Eq. (4.11)
10: Overwrite observed parcels using Eq. (4.12)
11: **return** $\hat{X}_{s,h}$

5 DLAM Atlas Prior Residual Variant

5.1 Intuition

The atlas-prior residual variant of DLAM keeps the subject-specific latent expression basis as the sole frame of representation, sidestepping the need for an independent atlas latent decomposition and the affine alignment between the two. The harmonized atlas is instead projected directly through the subject's gene loadings to give a dense atlas reference in the subject's own latent coordinates. The subject-specific deviation from this atlas reference is measured only at observed parcels and is propagated to unobserved parcels by a decaying-kernel radial-basis interpolator that vanishes far from the observed support, so predictions track the subject near data and relax to the atlas reference elsewhere. As in the original DLAM, harmonization, latent decomposition, residual interpolation, and decoding are estimated sequentially as point estimates.

5.2 Atlas projection and residual interpolation

The affine alignment in Eq. (4.5), the spatial transport in Eq. (4.7), the RBF on aligned spatial scores in Eq. (4.8), the ridge bridge in Eq. (4.9), and the inverse-affine in Eq. (4.10) are all replaced by a single subject-basis composition. The variant keeps the subject PLS of Eqs. (4.2)–(4.3) as the sole frame, projects the atlas reference through that same basis, and models the subject-specific deviation as a smooth spatial field with a decaying-kernel interpolator.

Atlas in the subject basis. The dense atlas is projected through the subject gene loadings,

$$Z_A^{(s)} \;=\; X_A^{(h)}\, W_s \;\in\; \mathbb{R}^{R\times K}, \tag{5.1}$$

yielding the atlas reference in the subject's own latent coordinates at every parcel. Because Z_s and $Z_A^{(s)}$ share the loadings W_s, both are decoded by the same map and no affine alignment is required.

Residual at observed parcels.

$$E_s \;=\; Z_s - Z_A^{(s)}[\Omega_s, :] \;\in\; \mathbb{R}^{|\Omega_s|\times K}. \tag{5.2}$$

Decaying-kernel residual interpolation. For each latent component $k = 1, \ldots, K$, an inverse-multiquadric (IMQ) RBF interpolator with length scale ℓ is fit on the anchors $\big(y_{\Omega_s,i},\, E_{s,i,k}\big)$:

$$\varphi(d) \;=\; \frac{1}{\sqrt{1 + (d/\ell)^2}}, \qquad d \geq 0. \tag{5.3}$$

Component-wise weights $\alpha_s^{(k)} \in \mathbb{R}^{|\Omega_s|}$ solve the regularized system

$$\big(\Phi_s + \lambda I\big)\, \alpha_s^{(k)} \;=\; E_{s,:,k}, \qquad (\Phi_s)_{ij} \;=\; \varphi\big(\|y_{\Omega_s,i} - y_{\Omega_s,j}\|\big), \tag{5.4}$$

and the field is evaluated at any parcel as

$$g_{s,k}(y_r) \;=\; \sum_{i=1}^{|\Omega_s|} \alpha_{s,i}^{(k)}\, \varphi\big(\|y_r - y_{\Omega_s,i}\|\big). \tag{5.5}$$

The IMQ kernel is strictly positive definite, monotonically decaying, and uses no polynomial tail, so $g_s(y_r) \to 0$ as $\min_i \|y_r - y_{\Omega_s,i}\| \to \infty$: the residual field vanishes far from observed parcels rather than extrapolating unboundedly.

Composition, decoding, and overwrite.

$$\hat{Z}_{s,r} \;=\; Z_A^{(s)}(r) \;+\; g_s(y_r), \qquad r = 1, \ldots, R, \tag{5.6}$$

$$\hat{X}_{s,h} \;=\; \hat{Z}_s\, W_s^{\top}. \tag{5.7}$$

Observed parcels are preserved by overwrite as in Eq. (4.12).

Behavioral envelope. Near observed parcels $g_s(y_r) \approx e_{s,r,:}$ and $\hat{Z}_{s,r} \to Z_s$, so the prediction tracks the subject. Off-support $g_s(y_r) \to 0$ and $\hat{Z}_{s,r} \to Z_A^{(s)}(r)$, so the prediction is the atlas reference in the subject's frame—bounded and decoded by the same W_s that decodes Z_s. Predictions therefore live between subject-near-data and an atlas-anchored fallback at all parcels, with no unbounded extrapolation regime.

5.3 Initialization

The shared harmonizer in Eq. (3.1) is fit first; the subject PLS in Eqs. (4.2)–(4.3) is fit on the harmonized observed matrix. The atlas projection in Eq. (5.1) and the residual in Eq. (5.2) are direct evaluations. The IMQ-RBF in Eqs. (5.4)–(5.5) is the only learned spatial map. No latent state is propagated across subjects. As an implementation safeguard, the per-gene standardization used to fit the subject PLS is shared with the atlas projection in Eq. (5.1) and lower-bounded by the atlas-side standard deviation, so out-of-sample projections of $X_A^{(h)}$ through a sparsely-supported basis remain stable on low-variance genes.

Algorithm 2 DLAM atlas-prior residual variant for subject s

Require: Atlas expression X_A, subject observations $X_s^{(G)}$ on Ω_s, parcel centroids Y, latent dimension K, IMQ length scale ℓ

Ensure: Completed harmonized atlas $\hat{X}_{s,h}$

1: Fit the shared harmonizer; compute $X_A^{(h)}$ and $X_{s,h}^{obs}$ via Eq. (3.1)
2: Fit the subject PLS in Eqs. (4.2)–(4.3) to obtain (W_s, Z_s)
3: Project the atlas through the subject basis: $Z_A^{(s)} = X_A^{(h)} W_s$ via Eq. (5.1)
4: Compute residual anchors E_s via Eq. (5.2)
5: Fit the IMQ-RBF residual interpolator via Eq. (5.4)
6: Evaluate $\hat{Z}_s$ at every parcel via Eqs. (5.5)–(5.6)
7: Decode to harmonized expression via Eq. (5.7)
8: Overwrite observed parcels via Eq. (4.12)
9: **return** $\hat{X}_{s,h}$

6 Probabilistic Latent Alignment Model (PLAM)

6.1 Intuition

PLAM takes the opposite view: instead of estimating a subject-specific basis first and then transporting it to the atlas, it treats the atlas as the primary latent geometry and represents each subject as a smooth, aligned deviation from that geometry. Subject-specific variation is captured by three components: an orthogonal alignment matrix, a spatially smooth latent deviation field, and subject-specific gene-wise calibration parameters. Unobserved parcels are inferred by conditioning the deviation field on the observed parcels and then decoding the completed latent state into harmonized expression.

6.2 Atlas factor model and subject-specific latent state

The harmonized atlas is represented by a low-rank factor model,

$$X_A^{(h)} \approx \mathbf{1}\alpha^\top + Z_A W^\top, \tag{6.1}$$

where $Z_A \in \mathbb{R}^{R\times K}$ contains atlas latent expression coordinates, $W \in \mathbb{R}^{G\times K}$ is a gene loading matrix, and $\alpha \in \mathbb{R}^G$ is a gene-wise intercept vector. This is the generative atlas model: each parcel's expression profile is explained by a K-dimensional latent code $z_{A,r}$ projected through the shared decoder W, plus a gene mean α. It is equivalent to probabilistic PCA with a fixed mean [Tipping and Bishop(1999)].

Let $z_{A,r}$ denote row r of Z_A and w_g denote row g of W. For subject s and parcel r, the latent state is

$$z_{s,r} = (z_{A,r} + \delta_{s,r})\, Q_s, \tag{6.2}$$

where $\delta_{s,r} \in \mathbb{R}^K$ is a subject-specific latent deviation from the atlas at parcel r, and $Q_s \in O(K)$ is an orthogonal alignment matrix. This decomposition cleanly separates three sources of variation: (i) the atlas coordinate $z_{A,r}$, shared across all subjects; (ii) the subject-specific deviation $\delta_{s,r}$, which is spatially smooth but subject-dependent; and (iii) the global rotation Q_s, which captures a linear reorientation of the subject's expression axes relative to the atlas. The corresponding platform-free latent expression is

$$x_{srg}^{(*)} = \alpha_g + z_{s,r}^\top w_g. \tag{6.3}$$

6.3 Subject-specific calibration

After the shared preprocessing harmonization in Eq. (3.1), residual platform mismatch may persist at the subject level. PLAM allows additional gene-wise subject-specific calibration in the observation model:

$$x_{srg}^{(G,h)} = a_{s,g}\, x_{srg}^{(*)} + b_{s,g} + \varepsilon_{srg}, \qquad r \in \Omega_s. \tag{6.4}$$

This calibration layer is structurally equivalent to the location-scale correction in ComBat [Johnson et al.(2007)], applied at the individual subject level within the latent observation model rather than at the cohort level in gene space. While the global harmonizer in Section 3.2 corrects for systematic platform differences shared across all subjects, the coefficients $(a_{s,g}, b_{s,g})$ absorb residual subject-specific offsets that remain after global correction.

Given the current latent states, the calibration parameters are updated by regularized regression,

$$\min_{a_{s,g},\, b_{s,g}} \sum_{r\in\Omega_s} \left(x_{srg}^{(G,h)} - a_{s,g}\tilde{x}_{srg}^{(*)} - b_{s,g} \right)^2 + \lambda_a (a_{s,g} - a_{0,g})^2 + \lambda_b (b_{s,g} - b_{0,g})^2, \tag{6.5}$$

where $\tilde{x}_{srg}^{(*)} = \alpha_g + z_{s,r}^\top w_g$ and $(a_{0,g}, b_{0,g})$ are global calibration targets. In the neutral initialization, $a_{0,g} = 1$ and $b_{0,g} = 0$, corresponding to the prior belief that the global harmonizer has already removed most platform effects.

6.4 Full joint distribution

The complete PLAM model is summarized by the following factored joint distribution:

$$\begin{aligned}
&p\Big(X_A^{(h)}, \{X_s^{(G,h)}\}_{s=1}^S, Z_A, W, \alpha, \{\Delta_s, Q_s, a_s, b_s\}_{s=1}^S\Big) \\
&\quad = p(X_A^{(h)} \mid Z_A, W, \alpha)\, p(Z_A)\, p(W)\, p(\alpha) \\
&\qquad \times \prod_{s=1}^S p(Q_s)\, p(\Delta_s \mid Y)\, p(a_s, b_s)\, p(X_s^{(G,h)} \mid Z_A, \Delta_s, Q_s, W, \alpha, a_s, b_s).
\end{aligned} \tag{6.6}$$

The explicit distributional assumptions for each factor are as follows.

Atlas likelihood.

$$p(X_A^{(h)} \mid Z_A, W, \alpha) = \prod_{r=1}^R \prod_{g=1}^G \mathcal{N}\Big(x_{A,rg}^{(h)};\ \alpha_g + z_{A,r}^\top w_g,\ \sigma_A^2\Big).$$

This is a Gaussian factor model for the atlas, identical to probabilistic PCA [Tipping and Bishop(1999)] with a fixed intercept α.

Atlas latent prior.

$$p(Z_A) = \prod_{r=1}^R \mathcal{N}(z_{A,r};\, \mathbf{0},\, I_K).$$

A standard spherical Gaussian prior on atlas latent codes.

Decoder and intercept priors.

$$p(W) = \prod_{g=1}^{G} \mathcal{N}(w_g;\, \mathbf{0},\, \tau_W^2 I_K), \qquad p(\alpha) = \prod_{g=1}^{G} \mathcal{N}(\alpha_g;\, 0,\, \tau_\alpha^2).$$

Gaussian shrinkage on gene loadings and gene means.

Orthogonal alignment.

$$p(Q_s) = \mathrm{Uniform}(O(K))\,.$$

Q_s is constrained to the group of $K \times K$ orthogonal matrices and estimated by solving a Procrustes problem (Section 6.6).

Spatial GP prior on deviations.

$$p(\Delta_s \mid Y) = \prod_{k=1}^{K} \mathcal{GP}(\delta_{s,\cdot,k};\; 0,\; \kappa_k(\cdot,\cdot))\,,$$

where κ_k is a squared-exponential kernel defined over parcel centroids (Section 6.7). This prior encodes the assumption that subject-specific deviations from the atlas are spatially smooth.

Calibration prior.

$$p(a_s, b_s) = \prod_{g=1}^{G} \mathcal{N}(a_{s,g};\, a_{0,g},\, \lambda_a^{-1})\,\mathcal{N}(b_{s,g};\, b_{0,g},\, \lambda_b^{-1}).$$

Gaussian regularization toward the identity transformation ($a_{0,g} = 1$, $b_{0,g} = 0$).

Subject likelihood.

$$p(X_s^{(G,h)} \mid \cdots) = \prod_{r \in \Omega_s} \prod_{g=1}^{G} \mathcal{N}\Big(x_{srg}^{(G,h)};\; a_{s,g}\big(\alpha_g + z_{s,r}^\top w_g\big) + b_{s,g},\; \sigma_{s,g}^2\Big)\,,$$

where $z_{s,r} = (z_{A,r} + \delta_{s,r})Q_s$ from Eq. (6.2).

6.5 From unweighted to weighted latent regression

The latent-state update is easiest to motivate from ordinary least squares. If every gene were treated as equally reliable and all residuals were penalized equally, the subject-specific latent coordinate at parcel r would solve

$$\min_z \sum_{g=1}^{G} \Big(x_{srg}^{(G,h)} - a_{s,g}\left(\alpha_g + z^\top w_g\right) - b_{s,g}\Big)^2 + \lambda_z \left\| z - z_{A,r} Q_s \right\|_2^2\,. \tag{6.7}$$

This objective balances fitting the observed gene expression against staying close to the atlas-aligned coordinate $z_{A,r}Q_s$.

In practice, some residuals are unusually large and some genes are systematically noisier than others. Let

$$e_{srg} = x_{srg}^{(G,h)} - a_{s,g}\left(\alpha_g + z_{s,r}^\top w_g\right) - b_{s,g}$$

denote the current residual. A Student-t-inspired robust weight is

$$\omega_{srg}^{robust} = \frac{\nu + 1}{\nu + e_{srg}^2 / \hat{\sigma}_{s,g}^2}, \tag{6.8}$$

and a gene-wise heteroscedastic weight is

$$\omega_{sg}^{hetero} = \left(\hat{\sigma}_{s,g}^2 + \epsilon_\omega\right)^{-1}, \qquad \hat{\sigma}_{s,g}^2 = \frac{1}{|\Omega_s|} \sum_{r \in \Omega_s} e_{srg}^2. \tag{6.9}$$

The combined weight is

$$\omega_{srg} = \omega_{srg}^{robust}\, \omega_{sg}^{hetero}. \tag{6.10}$$

The implemented latent-state update is then the weighted ridge regression

$$\min_z \sum_{g=1}^{G} \omega_{srg} \left(x_{srg}^{(G,h)} - a_{s,g} \left(\alpha_g + z^\top w_g \right) - b_{s,g} \right)^2 + \lambda_z \left\| z - z_{A,r} Q_s \right\|_2^2. \tag{6.11}$$

Connection to M-estimation and IRLS. The combined weight ω_{srg} corresponds to M-estimation [Huber(1981)]: ω_{srg}^{robust} is proportional to the influence function of a Student-t distribution, downweighting outlier genes at individual observations, while ω_{sg}^{hetero} accounts for gene-level heteroscedasticity. Fixing ω_{srg} and solving Eq. (6.11), then updating ω_{srg} given the new $z_{s,r}$, is precisely the iteratively reweighted least squares (IRLS) algorithm [Huber(1981)]. Each cycle is guaranteed not to increase the M-estimation objective.

6.6 Orthogonal Procrustes alignment

Given the updated observed latent coordinates, the orthogonal alignment matrix Q_s is refreshed by solving the classical orthogonal Procrustes problem [Schönemann(1966)]:

$$Q_s^\star = \arg \min_{Q \in O(K)} \left\| Z_{A,\Omega_s} Q - Z_{s,\Omega_s} \right\|_F^2. \tag{6.12}$$

This has the closed-form solution $Q_s^\star = UV^\top$, where $Z_{A,\Omega_s}^\top Z_{s,\Omega_s} = U \Sigma V^\top$ is the singular value decomposition.

6.7 Gaussian-process completion of the latent deviation field

The Gaussian process is used to interpolate the subject-specific latent deviation field rather than the expression directly [Rasmussen and Williams(2006)]. After the observed latent coordinates have been updated, the deviation at observed parcels is obtained by undoing the orthogonal alignment and subtracting the atlas coordinate:

$$\delta_{s,r}^{obs} = z_{s,r} Q_s^\top - z_{A,r}, \qquad r \in \Omega_s. \tag{6.13}$$

For latent dimension k, the scalar deviations $\{\delta_{s,r,k}^{obs}\}_{r \in \Omega_s}$ are modeled as noise-corrupted observations from a zero-mean Gaussian process with squared-exponential kernel

$$\kappa_k(y_r, y_{r'}) = \sigma_k^2 \exp\left(- \frac{\|y_r - y_{r'}\|_2^2}{2\ell_k^2} \right), \tag{6.14}$$

where σ_k^2 is the marginal signal variance and ℓ_k is the spatial length-scale for dimension k. Let $K_{\Omega,\Omega}^{(k)}$ denote the $|\Omega_s| \times |\Omega_s|$ kernel matrix evaluated at observed parcel centroids, $K_{r,\Omega}^{(k)}$ the $1 \times |\Omega_s|$ cross-kernel between parcel r and the observed parcels, and $K_{r,r}^{(k)}$ the prior variance at r. The GP posterior mean and variance at any parcel r are then

$$\hat{\delta}_{s,r,k} = K_{r,\Omega}^{(k)} \left(K_{\Omega,\Omega}^{(k)} + \tau_k^2 I \right)^{-1} \delta_{s,\Omega,k}^{obs}, \tag{6.15}$$

$$\mathrm{Var}(\delta_{s,r,k} \mid \Omega_s) = K_{r,r}^{(k)} - K_{r,\Omega}^{(k)} \left(K_{\Omega,\Omega}^{(k)} + \tau_k^2 I \right)^{-1} K_{\Omega,r}^{(k)}, \tag{6.16}$$

where τ_k^2 is an observation noise variance [Rasmussen and Williams(2006)]. The posterior mean $\hat{\delta}_{s,r,k}$ provides the completed deviation at unobserved parcel r, while the posterior variance quantifies how uncertain that completion is as a function of distance from the observed parcels.

Averaging the posterior variance over latent dimensions gives the parcel-wise average latent posterior variance,

$$\bar{\sigma}_{s,r}^2 = \frac{1}{K} \sum_{k=1}^{K} \mathrm{Var}(\delta_{s,r,k} \mid \Omega_s)\,. \tag{6.17}$$

The completed latent coordinate is reconstructed as

$$\hat{z}_{s,r} = \left(z_{A,r} + \hat{\delta}_{s,r} \right) Q_s. \tag{6.18}$$

The completed harmonized expression is decoded as

$$\hat{x}_{srg}^{(h)} = a_{s,g} \left(\alpha_g + \hat{z}_{s,r}^{\top} w_g \right) + b_{s,g}, \tag{6.19}$$

and observed parcels are preserved exactly by

$$\hat{x}_{srg}^{(h)} = x_{srg}^{(G,h)}, \qquad r \in \Omega_s. \tag{6.20}$$

6.8 Optional uncertainty shrinkage

The broader probabilistic family also permits the completed subject atlas to be shrunk toward an atlas prior according to spatial distance and average latent posterior variance. If $\tilde{d}_{s,r}$ is a scaled distance to the nearest observed parcel and $\tilde{\bar{\sigma}}_{s,r}^2$ is the scaled average latent posterior variance from Eq. (6.17), then one possible shrinkage score is

$$\mathrm{score}_{s,r} = \gamma_d \tilde{d}_{s,r} + \gamma_u \tilde{\bar{\sigma}}_{s,r}^2,$$

with logistic weight

$$w_{s,r} = \sigma\left(\frac{\mathrm{score}_{s,r} - m_0}{\tau} \right), \qquad \hat{x}_{s,r}^{final} = (1 - w_{s,r}) \hat{x}_{s,r}^{model} + w_{s,r} \hat{x}_r^{prior}.$$

This extension belongs to the general model family but is inactive in the particular probabilistic analysis presented here.

6.9 Initialization

Initialization is explicit because PLAM is fit by alternating updates. After the shared preprocessing harmonization in Eq. (3.1), the atlas latent coordinates Z_A are initialized by a rank-K singular value decomposition of centered $X_A^{(h)}$, and the decoder parameters (W, α) are fit by ridge regression under Eq. (6.1). For each subject, the orthogonal alignment is initialized as $Q_s = I$, the observed latent coordinates are initialized as the atlas coordinates at the observed parcels, $z_{s,r} = z_{A,r}$ for $r \in \Omega_s$, the calibration parameters are initialized as $a_{s,g} = 1$ and $b_{s,g} = 0$, and the combined weights are initialized as $\omega_{srg} = 1$. These choices correspond to the neutral hypothesis that the subject initially coincides with the atlas in the harmonized domain.

Algorithm 3 Unified probabilistic alignment for subject s

Require: Atlas expression X_A, subject observations $X_s^{(G)}$ on Ω_s, parcel centroids Y, latent dimension K, number of iterations T
Ensure: Completed harmonized atlas $\hat{X}_{s,h}$ and parcel-wise average latent posterior variance $\bar{\sigma}^2_{s,r}$

```
Fit the shared harmonizer and compute X_A^(h) and X_{s,h}^obs using Eq. (3.1)
Initialize Z_A, W, and α under the atlas factor model in Eq. (6.1)
Initialize Q_s ← I, z_{s,r} ← z_{A,r} for r ∈ Ω_s, a_{s,g} ← 1, b_{s,g} ← 0, and ω_{srg} ← 1
for t = 1, . . . , T do
    for each gene g do
        Update (a_{s,g}, b_{s,g}) by solving the regularized regression in Eq. (6.5)
    end for
    for each observed parcel r ∈ Ω_s do
        Form residuals e_{srg} and update ω_{srg}^robust, ω_{sg}^hetero, and ω_{srg} using Eqs. (6.8)–(6.10)
        Update z_{s,r} by solving the weighted latent regression in Eq. (6.11)
    end for
    Update Q_s by the orthogonal Procrustes step in Eq. (6.12)
end for
Form observed deviations δ_{s,r}^obs using Eq. (6.13)
for each latent dimension k = 1, . . . , K do
    Compute the Gaussian-process posterior mean and variance using Eqs. (6.15) and (6.16)
end for
Compute σ̄²_{s,r} using Eq. (6.17) and the completed latent coordinates ẑ_{s,r} using Eq. (6.18)
Decode to harmonized expression using Eq. (6.19)
Overwrite observed parcels using Eq. (6.20)
return X̂_{s,h} and σ̄²_{s,r}
```

7 Atlas-Level Diagnostics

Atlas-level diagnostics are useful because they separate two questions: whether the harmonization and alignment steps place AHBA and GTEx in a comparable coordinate system, and whether the imputation step preserves parcel-level structure when sparse subject data are aggregated into a population atlas. Figures 1 and 2 show the progression from raw AHBA and sparse GTEx observations to harmonized and completed parcel-by-gene atlases. In both cases, the gap between the sparse observed GTEx pattern and the completed harmonized atlas is explicitly visible. Figures 3 and 6 provide the same diagnostics for the naive atlas-fill baseline, making it possible to compare learned alignment against direct atlas borrowing in the same harmonized domain.

Figure 1: Atlas-level heatmap diagnostic for DLAM. Panels show (A) AHBA raw, (B) sparse GTEx raw, (C) sparse GTEx after harmonization, (D) AHBA harmonized, (E) the DLAM-completed harmonized GTEx atlas, and (F) the residual difference relative to the harmonized AHBA atlas.

Figure 2: Atlas-level heatmap diagnostic for PLAM. Panels show (A) AHBA raw, (B) sparse GTEx raw, (C) sparse GTEx after harmonization, (D) AHBA harmonized, (E) the PLAM-completed harmonized GTEx atlas, and (F) the residual difference relative to the harmonized AHBA atlas.

Figure 3: Atlas-level heatmap diagnostic for the naive atlas-fill baseline. Panels show (A) AHBA raw, (B) sparse GTEx raw, (C) sparse GTEx after harmonization, (D) AHBA harmonized, (E) the naive-fill harmonized GTEx atlas, and (F) the residual difference relative to the harmonized AHBA atlas.

Figure 4: Atlas-level scatter diagnostics for DLAM. Panels show (A) parcel identity, (B) gene identity, (C) parcel-level correlation, and (D) gene-level correlation in the harmonized atlas space. Only the most informative parcel and gene subsets are individually color-coded, with the remaining points shown in gray for context.

Figure 5: Atlas-level scatter diagnostics for PLAM. Panels show (A) parcel identity, (B) gene identity, (C) parcel-level correlation, and (D) gene-level correlation in the harmonized atlas space. The same visual grammar is used as in the DLAM panel so that the effect of the two alignment strategies can be compared directly.

Figure 6: Atlas-level scatter diagnostics for the naive atlas-fill baseline. Panels show (A) parcel identity, (B) gene identity, (C) parcel-level correlation, and (D) gene-level correlation in the harmonized atlas space.

8 Single-Subject Diagnostics

Single-subject diagnostics illustrate how each model turns sparse observations into a completed atlas. The representative subject shown here is `GTEX-14ASI`, chosen because its subject-level PLAM performance lies close to the cohort median among subjects with at least eight observed parcels. This makes the visual example representative of typical behavior rather than an extreme case.

A DLAM B PLAM C Naive fill

Figure 7: Single-subject heatmap diagnostics for the representative subject `GTEX-14ASI`. Panels (A)–(C) compare the transition from sparse observations to the completed harmonized atlas under DLAM, PLAM, and naive fill in the same visual format.

A DLAM B PLAM C Naive fill

Figure 8: Single-subject scatter diagnostics in harmonized space for the representative subject. Panels (A)–(C) show parcel identity, gene identity, parcel-level correlation, and gene-level correlation for the completed subject atlas under DLAM, PLAM, and naive fill.

9 Validation Context

Validation is based on leave-one-region-out cross-validation at the subject level. For each subject and each observed parcel, the parcel is held out completely, all fitted quantities are recomputed without it, and the held-out gene vector is predicted from the remaining parcels. This protocol is strict: the held-out parcel is excluded from harmonization, latent fitting, spatial completion, and decoding. The resulting diagnostics therefore measure genuine out-of-sample regional generalization rather than reconstruction of inputs seen during fitting.

Naive baseline. Both alignment models are benchmarked against a naive atlas-fill baseline. For each held-out parcel r of subject s, the predicted expression is simply the corresponding row of the harmonized AHBA atlas:

$$\hat{x}_{srg}^{naive} = x_{A,rg}^{(h)}, \qquad g = 1, \ldots, G.$$

This baseline captures the information available purely from the atlas spatial prior without any subject-specific adaptation. Improvement over this baseline therefore reflects the additional benefit of fitting each model to the individual subject's observed parcels.

The leave-one-region-out diagnostics reported below are computed over the full gene set so that the manuscript-level evaluation is consistent with the main transcriptome-wide completion setting. These plots are diagnostic rather than exhaustive: the aim is to expose how error and correlation vary over subjects, parcels, and genes, not merely to report a single cohort-level score.

Figure 9: Subject-level leave-one-region-out diagnostics over the full gene set. Subjects are ordered by PLAM Pearson correlation. The panels show the subject-wise coverage indicator, subject-wise Pearson correlation, subject-wise RMSE, and cohort-level summary bars for DLAM, PLAM, and the naive AHBA-fill baseline.

Figure 10: Parcel-level leave-one-region-out diagnostics over the full gene set. The heatmaps summarize mean held-out correlation and RMSE by parcel across the canonical parcel set; rows with no GTEx holdout events are left blank. The bar plot shows parcel-specific improvement relative to the naive AHBA-fill baseline for parcels with available leave-one-region-out evaluations.

Figure 11: Cohort-level summary of leave-one-region-out diagnostics over the full gene set. Mean Pearson correlation, mean RMSE, and mean improvement relative to the naive AHBA-fill baseline are shown for both atlas-completion models and the naive baseline.

10 Discussion

DLAM and PLAM share a common scientific aim: to infer a subject-complete parcel-by-gene atlas from sparse GTEx regional measurements by borrowing structure from AHBA. Their main difference lies in what is treated as primary. DLAM treats each subject as the starting point, estimates a subject-specific latent basis whose spatial structure is revealed by PLS, and then transports that basis into the atlas frame. PLAM treats the atlas as the starting point, defines a global latent geometry there, and then expresses each subject as a smooth, orthogonally aligned perturbation of that geometry.

This distinction has practical consequences. DLAM is modular and interpretable at each step, which makes it attractive for diagnosing where alignment or extrapolation may fail. PLAM is more coherent because calibration, alignment, latent completion, and uncertainty all live in a single atlas-referenced latent system. That coherence is what makes Gaussian-process deviation conditioning natural in the probabilistic formulation: the GP is not interpolating gene expression directly, but a low-dimensional deviation field whose spatial smoothness is biologically and statistically motivated [Rasmussen and Williams(2006)].

The two approaches should therefore not be understood simply as different stages of a software pipeline. They represent two different answers to the same problem. DLAM provides a transparent latent transport view built from subject-specific components, whereas PLAM provides a global generative view in which atlas structure, subject alignment, subject deviation, and calibration are estimated jointly in a common latent space.

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